Metric Markov cotype two of ℓ_1 with constant 3024: an exposition of a machine-checked proof

Abstract

We prove that ℓ_1 has metric Markov cotype two with $C^2 = 3024$, that is, $N_2(\ell_1) \le 12\sqrt{21}$. For every stationary pair (π, A) on $[n]$, every $t \ge 1$ and every $x_1, \dots, x_n \in \ell_1$, there are points $y_i \in \ell_1$ whose combined fidelity and smoothness cost is at most 3024 times the Cesàro-averaged energy of the x_i. The argument has three parts. First, the points are encoded by weighted cut coordinates. Second, the cubic map $r \mapsto 3r^2 - 2r^3$ is applied coordinatewise to resolvent averages of the binary encodings, and its energy is bounded by a quartic potential that telescopes against the resolvent $p\sum_s q^s A^s$ with $p = 1/(t+1)$. Third, the geometric (resolvent) average of endpoint costs is compared with the Cesàro average, in the spirit of Mendel–Naor. The main theorem and all lemmas below are machine-verified. The probabilistic martingale of the source argument is replaced in the formal proof by an equivalent algebraic telescoping.

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1 Setting

Index sets and matrices. Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. For $n \in \mathbb{N}$ let $[n] = \{1, \dots, n\}$ with its natural order; $[n]$ is empty when $n = 0$. Vectors are indexed by $[n]$ and matrices by $[n] \times [n]$. For a matrix A we write $a_{ij} = A_{ij}$, we write δ_{ij} for the Kronecker delta, and we set $A^0 = I$ and

$A^{s+1} = A^s A$. We write $\mathcal{P}_n$ for the set of all 2^n subsets of $[n]$, including $\emptyset$ and $[n]$. For $i \in [n]$ the binary vector $\chi_i \in \{0,1\}^{\mathcal{P}_n}$ is defined by $\chi_i(B) = 1$ if $i \in B$ and $\chi_i(B) = 0$ otherwise.

For $s, t \in \mathbb{N}$ with $t \geq 1$, $\lfloor s/t \rfloor$ and $s \bmod t$ are the quotient and remainder of Euclidean division, so that $s = t\lfloor s/t \rfloor + (s \bmod t)$. For $t = 0$ we use the conventions $\lfloor s/0 \rfloor = 0$ and $s \bmod 0 = s$, and also $1/0 = 0$. These conventions only affect degenerate cases, which are pointed out where they occur.

Series. For a real sequence $(c_s)_{s \in \mathbb{N}}$ we write $\sum_{s \geq 0} c_s$ for the sum of the series when the series is summable (equivalently, absolutely convergent), and 0 otherwise. Whenever a lemma asserts convergence, it says so explicitly. In every place where this convention is used, the proof shows that the series converges.

Definition 1.1 (The space ℓ_1)**.** ℓ_1 is the real vector space of sequences $f = (f(k))_{k \in \mathbb{N}}$ with $\sum_k |f(k)|^p < \infty$ for $p = 1$. It carries pointwise operations and the norm $\|f\|_1 = \left(\sum_k |f(k)|^p\right)^{1/p}$ with $p = 1$. This makes ℓ_1 a real Banach space. Indexing by $\mathbb{N}$ instead of $\{1, 2, \dots\}$ is immaterial, since the index shift is a linear isometry onto.

Definition 1.2 (Probability vectors, stochastic and reversible matrices)**.** Let $n \in \mathbb{N}$.

- A vector $\pi \in \mathbb{R}^n$ is a *probability vector* if $\pi_i \geq 0$ for all i and $\sum_i \pi_i = 1$. Zero entries are allowed.
- A matrix $A \in \mathbb{R}^{n \times n}$ is *stochastic* if $a_{ij} \geq 0$ for all i, j and $\sum_j a_{ij} = 1$ for all i.
- A is *reversible* with respect to $\pi \in \mathbb{R}^n$ if $\pi_i a_{ij} = \pi_j a_{ji}$ for all i, j.
- π is *stationary* for A, written $\pi A = \pi$, if $\sum_i \pi_i a_{ij} = \pi_j$ for all j.

Resolvent parameters. For $t \in \mathbb{N}$ put

$$p = p_t = \frac{1}{t+1}, \qquad q = q_t = \frac{t}{t+1}.$$

Then $p > 0$, $0 \leq q < 1$, $p + q = 1$, $(t+1)p = 1$ and $(t+1)q = t$. For a stochastic matrix A, the *resolvent matrix* G is the matrix with entries

$$G_{ij} = \sum_{s \geq 0} p\, q^s (A^s)_{ij}.$$

These series converge by Lemma 2.4.

Weighted seminorms and the cubic map. Let ι be a finite set and $w \in \mathbb{R}^\iota$ with $w_b \geq 0$ for all b. For $u, v \in \mathbb{R}^\iota$ put

$$\|u\|_{w,1} = \sum_b w_b |u_b|, \qquad \langle u, v \rangle_w = \sum_b w_b u_b v_b, \qquad \|u\|_w^2 = \langle u, u \rangle_w, \qquad \|u\|_w^4 = (\|u\|_w^2)^2.$$

Let $\phi(r) = 3r^2 - 2r^3$ for $r \in \mathbb{R}$. For $u \in \mathbb{R}^\iota$ let $\Phi(u) \in \mathbb{R}^\iota$ be defined by $\Phi(u)_b = \phi(u_b)$. We write $Q = [0,1]^\iota$ for the cube and $\{0,1\}^\iota$ for its set of vertices.

Theorem 1.3 (Metric Markov cotype two of ℓ_1)**.** *Let $n, t \in \mathbb{N}$ with $t \geq 1$. Let π be a probability vector on $[n]$, let A be a stochastic matrix on $[n]$ that is reversible with respect to π, and let $x_1, \ldots, x_n \in \ell_1$. Then there exist $y_1, \ldots, y_n \in \ell_1$ such that*

$$\sum_i \pi_i \|x_i - y_i\|_1^2 + t \sum_{i,j} \pi_i a_{ij} \|y_i - y_j\|_1^2 \leq 3024 \sum_{i,j} \pi_i \Big(\frac{1}{t} \sum_{s=1}^{t} (A^s)_{ij} \Big) \|x_i - x_j\|_1^2.$$

This is exactly the statement of the source, with $C^2 = 3024$, so that $N_2(\ell_1) \leq \sqrt{3024} = 12\sqrt{21}$. The only formal difference is that sequences in ℓ_1 are indexed from 0, which is an isometric relabelling. As in the source, reversibility enters only through stationarity.

2 Lemmas

2.1 Stationarity and powers of stochastic matrices

Lemma 2.1 (Reversibility implies stationarity)**.** *Let $\pi \in \mathbb{R}^n$ and let A be stochastic and reversible with respect to π. Then $\sum_i \pi_i a_{ij} = \pi_j$ for every j.*

Proof. By detailed balance and the row-sum condition,

$$\sum_i \pi_i a_{ij} = \sum_i \pi_j a_{ji} = \pi_j \sum_i a_{ji} = \pi_j. \qquad \square$$

Lemma 2.2 (Powers of stochastic matrices)**.** *If A is stochastic, then A^s is stochastic for every $s \in \mathbb{N}$.*

Proof. We argue by induction on s. The identity matrix is stochastic. If A^s is stochastic, then $(A^{s+1})_{ij} = \sum_k (A^s)_{ik} a_{kj} \geq 0$. Moreover

$$\sum_j \sum_k (A^s)_{ik} a_{kj} = \sum_k (A^s)_{ik} \sum_j a_{kj} = \sum_k (A^s)_{ik} = 1. \qquad \square$$

Lemma 2.3 (Stationarity of powers)**.** *Let $\pi \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$ satisfy $\pi A = \pi$. Then $\sum_i \pi_i (A^s)_{ij} = \pi_j$ for all $s \in \mathbb{N}$ and all j.*

Proof. We argue by induction on s; the case $s = 0$ is trivial. Using $A^{s+1} = A^s A$, exchanging finite sums and applying the induction hypothesis,

$$\sum_i \pi_i \sum_k (A^s)_{ik} a_{kj} = \sum_k \Big(\sum_i \pi_i (A^s)_{ik} \Big) a_{kj} = \sum_k \pi_k a_{kj} = \pi_j. \qquad \square$$

2.2 The resolvent

Lemma 2.4 (Resolvent series converge)**.** *Let $t \in \mathbb{N}$, let A be stochastic and let $i, j \in [n]$. Then the series $\sum_{s \geq 0} p\, q^s (A^s)_{ij}$ is summable.*

Proof. First, $0 \leq (A^s)_{ij} \leq 1$ for all s. This follows by induction from $(A^{s+1})_{ij} = \sum_k a_{ik} (A^s)_{kj}$: the sum is nonnegative, and it is at most $\sum_k a_{ik} = 1$. Hence $0 \leq p\, q^s (A^s)_{ij} \leq p\, q^s$. Since $0 \leq q < 1$, the geometric series $\sum_s p\, q^s$ converges, and the claim follows by comparison. $\square$

Lemma 2.5 (Basic properties of the resolvent)**.** *Let $t \in \mathbb{N}$ and $\pi \in \mathbb{R}^n$. Let A be stochastic with $\pi A = \pi$, and let G be the resolvent matrix, $G_{ij} = \sum_{s \geq 0} p\, q^s (A^s)_{ij}$. Then:*

1. $G_{ij} \geq 0$ *for all* i, j;
2. $\sum_j G_{ij} = 1$ *for all* i;
3. $\sum_i \pi_i G_{ij} = \pi_j$ *for all* j.

Proof. Since $1 - q = p$, we have $\sum_{s \geq 0} p\, q^s = p/(1-q) = 1$. All series $\sum_s p\, q^s (A^s)_{ij}$ converge by Lemma 2.4, so finite sums and constant multiples may be taken termwise.

1. Each term is nonnegative by Lemma 2.2.
2. By Lemma 2.2, $\sum_j G_{ij} = \sum_s p\, q^s \sum_j (A^s)_{ij} = \sum_s p\, q^s = 1$.
3. By Lemma 2.3, $\sum_i \pi_i G_{ij} = \sum_s p\, q^s \sum_i \pi_i (A^s)_{ij} = \sum_s p\, q^s \pi_j = \pi_j$. □

Lemma 2.6 (First-step identities)**.** *Let* $t \in \mathbb{N}$*, let* A *be stochastic and let* G *be its resolvent matrix. Then for all* i, j*,*

$$G_{ij} = p\, \delta_{ij} + q \sum_k a_{ik} G_{kj} \qquad \text{and} \qquad G_{ij} = p\, \delta_{ij} + q \sum_k G_{ik} a_{kj}.$$

Proof. All series involved converge by Lemma 2.4. Hence

$$\sum_k a_{ik} G_{kj} = \sum_{s \geq 0} p\, q^s \sum_k a_{ik} (A^s)_{kj} = \sum_{s \geq 0} p\, q^s (A^{s+1})_{ij}.$$

Splitting off the term $s = 0$ of G_{ij}, which equals $p\, \delta_{ij}$, gives

$$G_{ij} = p\, \delta_{ij} + \sum_{s \geq 0} p\, q^{s+1} (A^{s+1})_{ij} = p\, \delta_{ij} + q \sum_k a_{ik} G_{kj}.$$

The second identity follows in the same way from $A^{s+1} = A^s A$. □

Lemma 2.7 (Resolvent averages of binary vectors)**.** *Let* $t \in \mathbb{N}$ *and* $A, G \in \mathbb{R}^{n \times n}$*. Assume* $G_{ij} \geq 0$*,* $\sum_j G_{ij} = 1$ *and* $G_{ij} = p\, \delta_{ij} + q \sum_k a_{ik} G_{kj}$ *for all* i, j*. Let* ι *be any set and let* $z_1, \ldots, z_n : \iota \to \{0, 1\}$*. Put* $h_i(b) = \sum_j G_{ij} z_j(b)$*. Then for all* $i \in [n]$ *and* $b \in \iota$*:*

1. $0 \leq h_i(b) \leq 1$;
2. $h_i(b) = p\, z_i(b) + q \sum_k a_{ik} h_k(b)$.

Proof. Part (1) holds because $h_i(b)$ is a convex combination of numbers in $\{0, 1\}$. For part (2), multiply the hypothesis $G_{ij} = p\, \delta_{ij} + q \sum_k a_{ik} G_{kj}$ by $z_j(b)$, sum over j, and exchange the finite sums over j and k:

$$h_i(b) = p\, z_i(b) + q \sum_k a_{ik} \sum_j G_{kj} z_j(b).$$ □

2.3 The cubic map and the quartic potential

Lemma 2.8 (Scalar increment bound)**.** *For* $r, r' \in [0, 1]$ *and* $d \in \{0, 1\}$*,*

$$|\phi(r') - \phi(r)| \leq 6|r - d|\, |r' - r| + 3(r' - r)^2.$$

Proof. Let $\delta = r' - r$. Factoring the difference of squares and of cubes gives

$$\phi(r') - \phi(r) = \delta\, K, \qquad K = 3(r' + r) - 2\big(r'^2 + r'r + r^2\big).$$

It therefore suffices to show

$$|K| \le 6|r - d| + 3|\delta|,$$

since multiplying by $|\delta|$ then gives the claim. In the formal proof this bound is obtained by case analysis on $d \in \{0, 1\}$ and on the sign of δ; in each case it is a polynomial inequality for $(r, r') \in [0, 1]^2$, discharged by nonlinear arithmetic. A direct way to see it is as follows. Substituting $r' = r + \delta$ gives $K = 6r(1 - r) + \delta(3 - 4r - 2r')$.

- Since $4r + 2r' \in [0, 6]$, we have $|3 - 4r - 2r'| \le 3$.
- We have $0 \le 6r(1 - r) \le 6|r - d|$. If $d = 0$, then $|r - d| = r$ and $1 - r \le 1$. If $d = 1$, then $|r - d| = 1 - r$ and $r \le 1$.

The triangle inequality gives the bound on $|K|$. □

Lemma 2.9 (Weighted increment bound)**.** *Let ι be finite, $w \in \mathbb{R}^\iota$ with $w \ge 0$, $u, u' \in [0, 1]^\iota$ and $e \in \{0, 1\}^\iota$. Then*

$$\|\Phi(u') - \Phi(u)\|_{w,1}^2 \le 72\, \|u - e\|_w^2\, \|u' - u\|_w^2 + 18\, \|u' - u\|_w^4.$$

Proof. Put

$$X = \|\Phi(u') - \Phi(u)\|_{w,1}, \qquad S = \sum_b w_b |u_b - e_b|\, |u'_b - u_b|, \qquad P = \|u' - u\|_w^2.$$

Linear bound. The formal proof does not cite Lemma 2.8. It re-establishes the same scalar bound inline: for $r, r' \in [0, 1]$ and $d \in \{0, 1\}$,

$$|\phi(r') - \phi(r)| \le 6|r - d|\, |r' - r| + 3(r' - r)^2.$$

This inline argument uses the expansion $\phi(r') - \phi(r) = 6r(1 - r)\,\delta + \delta^2(3 - 6r - 2\delta)$ with $\delta = r' - r$, together with the estimates $|3 - 6r - 2\delta| = |3 - 4r - 2r'| \le 3$ and $0 \le 6r(1 - r) \le 6|r - d|$. Applying this bound in each coordinate with $d = e_b$, multiplying by $w_b \ge 0$ and summing gives $0 \le X \le 6S + 3P$.

Cauchy–Schwarz. Writing $w_b = (\sqrt{w_b})^2$, the Cauchy–Schwarz inequality gives $S^2 \le \|u - e\|_w^2 P$.

Squaring. From $(6S - 3P)^2 \ge 0$ we get $36SP \le 36S^2 + 9P^2$. Therefore

$$X^2 \le (6S + 3P)^2 \le 72S^2 + 18P^2 \le 72\, \|u - e\|_w^2 P + 18P^2.$$ □

Lemma 2.10 (Flatness against the quartic potential)**.** *Under the hypotheses of Lemma 2.9,*

$$\|\Phi(u') - \Phi(u)\|_{w,1}^2 \le 108\Big(\|u' - e\|_w^4 - \|u - e\|_w^4 - 4\, \|u - e\|_w^2\, \langle u - e, u' - u\rangle_w\Big).$$

Proof. Put $S = \|u - e\|_w^2$, $P = \|u' - u\|_w^2$ and $I = \langle u - e, u' - u\rangle_w$. Then $\|u' - e\|_w^2 = S + 2I + P$, so the bracket equals

$$R = (S + 2I + P)^2 - S^2 - 4SI = 2SP + (2I + P)^2.$$

By Cauchy–Schwarz (again with $\sqrt{w_b}$), $I^2 \le SP$. Hence

$$P^2 \le 2(2I + P)^2 + 2(2I)^2 \le 2(2I + P)^2 + 8SP.$$

Combining this with Lemma 2.9,

$$\|\Phi(u') - \Phi(u)\|_{w,1}^2 \le 72SP + 18P^2 \le 216SP + 36(2I + P)^2 \le 108R.$$ □

Lemma 2.11 (One averaged step)**.** *Let ι be finite, $w \in \mathbb{R}^\iota$ with $w \geq 0$ and $e \in \{0,1\}^\iota$. Let $p, q \geq 0$ with $p + q = 1$, and let $a \in \mathbb{R}^n$ with $a_k \geq 0$ and $\sum_k a_k = 1$. Let $u, v, h_1, \ldots, h_n \in [0,1]^\iota$ satisfy*

$$u_b = p\, v_b + q \sum_k a_k h_k(b) \quad \text{for all } b.$$

Then

$$p\, \|\Phi(v) - \Phi(u)\|_{w,1}^2 + q \sum_k a_k \|\Phi(h_k) - \Phi(u)\|_{w,1}^2 \\ \leq 108\Big(p\, \|v - e\|_w^4 + q \sum_k a_k \|h_k - e\|_w^4 - \|u - e\|_w^4\Big).$$

Proof. Apply Lemma 2.10 to the pair (u, v) and to each pair (u, h_k). Multiply these inequalities by p and by $q\, a_k$ respectively, which are nonnegative, and add them.

Constant terms. Since $p + q \sum_k a_k = 1$, the terms $-\|u - e\|_w^4$ combine to $-\|u - e\|_w^4$.

Linear terms. These carry the common factor $-4\|u - e\|_w^2$ and the cross term

$$p\, \langle u - e, v - u\rangle_w + q \sum_k a_k \langle u - e, h_k - u\rangle_w = \sum_b w_b (u_b - e_b)\Big(p\, v_b + q \sum_k a_k h_k(b) - (p+q) u_b\Big).$$

This vanishes by the hypothesis on u and $p + q = 1$. □

Lemma 2.12 (Aggregation against the resolvent)**.** *Let $t \in \mathbb{N}$ and $\pi \in \mathbb{R}^n$ with $\pi_i \geq 0$. Let $A, G \in \mathbb{R}^{n \times n}$ satisfy:*

- *$G_{ij} \geq 0$ and $G_{ij} = p\, \delta_{ij} + q \sum_k G_{ik} a_{kj}$ for all i, j;*
- *$\sum_i \pi_i G_{ij} = \pi_j$ for all j.*

Let $\beta \in \mathbb{R}^n$ and $E, F, H \in \mathbb{R}^{n \times n}$ with $H_{ij} \geq 0$ for all i, j. Assume that for all $i, j \in [n]$,

$$p\, \beta_j + q \sum_k a_{jk} E_{jk} \leq 108\Big(p\, F_{ij} + q \sum_k a_{jk} H_{ik} - H_{ij}\Big).$$

Then

$$\sum_j \pi_j \beta_j + t \sum_j \sum_k \pi_j a_{jk} E_{jk} \leq 108 \sum_i \sum_j \pi_i G_{ij} F_{ij}.$$

Proof. Multiply the hypothesis at (i, j) by $\pi_i G_{ij} \geq 0$ and sum over i and j.

Left side. Using $\sum_i \pi_i G_{ij} = \pi_j$, it becomes

$$\sum_j \pi_j \Big(p \beta_j + q \sum_k a_{jk} E_{jk}\Big).$$

Right side. Fix i. By the hypothesis on G, $q \sum_j G_{ij} a_{jk} = G_{ik} - p\, \delta_{ik}$. Therefore

$$\sum_j G_{ij}\, q \sum_k a_{jk} H_{ik} = \sum_k (G_{ik} - p\, \delta_{ik}) H_{ik} = \sum_k G_{ik} H_{ik} - p\, H_{ii}.$$

The i-th contribution to the right side is thus

$$108\, \pi_i \Big(p \sum_j G_{ij} F_{ij} - p\, H_{ii}\Big) \leq 108\, p\, \pi_i \sum_j G_{ij} F_{ij},$$

since $\pi_i p\, H_{ii} \geq 0$.

Conclusion. Summing over i gives

$$p \sum_j \pi_j \beta_j + q \sum_{j,k} \pi_j a_{jk} E_{jk} \leq 108\, p \sum_{i,j} \pi_i G_{ij} F_{ij}.$$

Multiplying by $t+1 > 0$ and using $(t+1)p = 1$ and $(t+1)q = t$ gives the claim. □

Lemma 2.13 (Vertices are fixed)**.** *If $d \in \{0,1\}$, then $\phi(d) = d$.*

Proof. $\phi(0) = 0$ and $\phi(1) = 3 - 2 = 1$. □

Lemma 2.14 (One step for the resolvent averages)**.** *Let $t \in \mathbb{N}$ and let A be stochastic. Let ι be finite, $w \in \mathbb{R}^\iota$ with $w \geq 0$, and $z_1, \ldots, z_n \in \{0,1\}^\iota$. Let $G \in \mathbb{R}^{n \times n}$ satisfy $G_{ij} \geq 0$, $\sum_j G_{ij} = 1$ and $G_{ij} = p\,\delta_{ij} + q \sum_k a_{ik} G_{kj}$ for all i, j. Put $h_k = \sum_l G_{kl} z_l \in \mathbb{R}^\iota$. Then for all $i_0, j \in [n]$,*

$$p\, \|\Phi(z_j) - \Phi(h_j)\|_{w,1}^2 + q \sum_k a_{jk} \|\Phi(h_k) - \Phi(h_j)\|_{w,1}^2 \\ \leq 108 \Big(p\, \|z_j - z_{i_0}\|_w^4 + q \sum_k a_{jk} \|h_k - z_{i_0}\|_w^4 - \|h_j - z_{i_0}\|_w^4 \Big).$$

Proof. We first derive two facts directly from the hypotheses on G. These are the conclusions of Lemma 2.7, but the formal proof re-derives them rather than citing it.

- $h_k \in [0,1]^\iota$ *for all* k. For each b, $h_k(b) = \sum_l G_{kl} z_l(b)$ with $G_{kl} \geq 0$, $\sum_l G_{kl} = 1$ and $z_l(b) \in \{0,1\}$. Hence $0 \leq h_k(b) \leq \sum_l G_{kl} = 1$.
- *The first-step identity* $h_j = p\, z_j + q \sum_k a_{jk} h_k$. Multiply $G_{jl} = p\,\delta_{jl} + q \sum_k a_{jk} G_{kl}$ by $z_l(b)$, sum over l, and exchange the finite sums over l and k.

Moreover $z_j \in \{0,1\}^\iota \subseteq [0,1]^\iota$, $p, q \geq 0$ and $p + q = 1$, and the row $(a_{jk})_k$ is nonnegative with sum 1. Apply Lemma 2.11 with $e = z_{i_0}$, $a_k = a_{jk}$, $u = h_j$, $v = z_j$ and the vectors h_k. □

Lemma 2.15 (Energy bound, raw form)**.** *Let $t \in \mathbb{N}$, let $\pi \in \mathbb{R}^n$ with $\pi_i \geq 0$ and let A be stochastic. Let ι be finite, $w \in \mathbb{R}^\iota$ with $w \geq 0$, and $z_1, \ldots, z_n \in \{0,1\}^\iota$. Let $G \in \mathbb{R}^{n \times n}$ satisfy:*

- $G_{ij} \geq 0$ *and* $\sum_j G_{ij} = 1$*;*
- *both identities* $G_{ij} = p\,\delta_{ij} + q \sum_k a_{ik} G_{kj}$ *and* $G_{ij} = p\,\delta_{ij} + q \sum_k G_{ik} a_{kj}$*;*
- $\sum_i \pi_i G_{ij} = \pi_j$.

Let $h_k = \sum_l G_{kl} z_l$. Then

$$\sum_j \pi_j \|\Phi(z_j) - \Phi(h_j)\|_{w,1}^2 + t \sum_j \sum_k \pi_j a_{jk} \|\Phi(h_k) - \Phi(h_j)\|_{w,1}^2 \leq 108 \sum_i \sum_j \pi_i G_{ij} \|z_j - z_i\|_w^4.$$

Proof. Apply Lemma 2.12 with

$$\beta_j = \|\Phi(z_j) - \Phi(h_j)\|_{w,1}^2, \quad E_{jk} = \|\Phi(h_k) - \Phi(h_j)\|_{w,1}^2, \quad F_{ij} = \|z_j - z_i\|_w^4, \quad H_{ij} = \|h_j - z_i\|_w^4 \geq 0.$$

Its hypothesis at (i,j) is exactly Lemma 2.14 with $i_0 = i$. □

Lemma 2.16 (Energy bound)**.** *Assume the hypotheses of Lemma 2.15, except that π is now a probability vector. Then*

$$\sum_i \pi_i \|z_i - \Phi(h_i)\|_{w,1}^2 + t \sum_i \sum_j \pi_i a_{ij} \|\Phi(h_i) - \Phi(h_j)\|_{w,1}^2 \leq 108 \sum_i \sum_j \pi_i G_{ij} \|z_i - z_j\|_w^4.$$

Proof. This is Lemma 2.15, which only uses $\pi \geq 0$, rewritten in two ways. First, $\Phi(z_i) = z_i$ by Lemma 2.13. Second, the seminorms are invariant under $v \mapsto -v$. □

2.4 Cut decomposition in ℓ_1

Lemma 2.17 (Cuts of a monotone vector)**.** *Let $n \in \mathbb{N}$ and let $a \in \mathbb{R}^n$ be nondecreasing, that is, $i \le j$ implies $a_i \le a_j$. Then there exist subsets $c_1, \dots, c_n \subseteq [n]$ and reals $g_1, \dots, g_n, v_0$ such that:*

1. *$g_r \ge 0$ for all r;*
2. *$\sum_r g_r \le 2 \sum_i |a_i|$;*
3. *$|v_0| \le \sum_i |a_i|$;*
4. *$a_i = v_0 + \sum_r \mathbf{1}[i \in c_r]\, g_r$ for all i;*
5. *$|a_i - a_j| = \sum_r |\mathbf{1}[i \in c_r] - \mathbf{1}[j \in c_r]|\, g_r$ for all i, j.*

Proof. If $n = 0$, everything is empty and all claims are trivial. Otherwise define

$$c_r = \{i : i > r\}, \qquad g_r = a_{r+1} - a_r \ (r < n), \qquad g_n = 0, \qquad v_0 = a_1.$$

- Monotonicity gives $g_r \ge 0$.
- By telescoping, $\sum_{r<m} g_r = a_m - a_1$ for every $m \in [n]$.
- Since $\mathbf{1}[i \in c_r]\, g_r = \mathbf{1}[r < i]\, g_r$, the reconstruction (4) follows.
- We have $\sum_r g_r = a_n - a_1 \le |a_n| + |a_1| \le 2 \sum_i |a_i|$, and $|v_0| = |a_1| \le \sum_i |a_i|$.

For (5), let $i \le j$. A three-case check ($r < i$, then $i \le r < j$, then $r \ge j$) gives

$$|\mathbf{1}[i \in c_r] - \mathbf{1}[j \in c_r]| g_r = \mathbf{1}[r < j] g_r - \mathbf{1}[r < i] g_r.$$

Summing over r gives $(a_j - a_1) - (a_i - a_1) = |a_i - a_j|$. The case $i > j$ follows by symmetry. □

Lemma 2.18 (Cuts of an arbitrary vector)**.** *The conclusion of Lemma 2.17 holds for every $a \in \mathbb{R}^n$, without the monotonicity assumption.*

Proof. Choose a permutation σ of $[n]$ such that $a \circ \sigma$ is nondecreasing; such a σ exists by sorting. Apply Lemma 2.17 to $a \circ \sigma$, obtaining c'_r, g_r, v_0, and set $c_r = \sigma(c'_r)$. Then $\mathbf{1}[i \in c_r] = \mathbf{1}[\sigma^{-1}(i) \in c'_r]$. Hence (4) and (5) for a follow from those for $a \circ \sigma$ evaluated at $\sigma^{-1}(i)$ and $\sigma^{-1}(j)$. Finally, $\sum_i |a_{\sigma(i)}| = \sum_i |a_i|$. □

Lemma 2.19 (Uniform cut decomposition of $\mathbb{R}^n$)**.** *Let $n \in \mathbb{N}$. There exist functions $g : \mathcal{P}_n \times \mathbb{R}^n \to \mathbb{R}$, $(B, a) \mapsto g_B(a)$, and $v : \mathbb{R}^n \to \mathbb{R}$ such that for every $a \in \mathbb{R}^n$:*

1. *$0 \le g_B(a) \le 2 \sum_i |a_i|$ for all $B \in \mathcal{P}_n$;*
2. *$|v(a)| \le \sum_i |a_i|$;*
3. *$a_i = v(a) + \sum_{B \in \mathcal{P}_n} \mathbf{1}[i \in B]\, g_B(a)$ for all i;*
4. *$|a_i - a_j| = \sum_{B \in \mathcal{P}_n} |\mathbf{1}[i \in B] - \mathbf{1}[j \in B]|\, g_B(a)$ for all i, j.*

Proof. For each a choose data $c_r(a), g_r(a), v_0(a)$ as in Lemma 2.18. Put $v(a) = v_0(a)$ and

$$g_B(a) = \sum_{r:\, c_r(a)=B} g_r(a).$$

For any function f on $\mathcal{P}_n$ we have $\sum_B f(B)\, g_B(a) = \sum_r f(c_r(a))\, g_r(a)$. Applying this to $f(B) = \mathbf{1}[i \in B]$ and to $f(B) = |\mathbf{1}[i \in B] - \mathbf{1}[j \in B]|$ transfers (3) and (4) from Lemma 2.18. Finally, $0 \le g_B(a) \le \sum_r g_r(a) \le 2 \sum_i |a_i|$. □

Lemma 2.20 (The ℓ_1 norm)**.** *For $f \in \ell_1$, the sequence $(|f(k)|)_k$ is summable and $\|f\|_1 = \sum_k |f(k)|$.*

Proof. This is the definition of membership in ℓ_p and of the ℓ_p norm, specialised to $p = 1$. □

Lemma 2.21 (Domination gives membership in ℓ_1)**.** *Let $f, g : \mathbb{N} \to \mathbb{R}$ with g summable and $|f(k)| \le g(k)$ for all k. Then there exists $F \in \ell_1$ with $F(k) = f(k)$ for all k.*

Proof. By comparison, $\sum_k |f(k)| < \infty$, so f itself belongs to ℓ_1. □

Lemma 2.22 (Coordinates of affine combinations)**.** *Let ι be finite, $V \in \ell_1$, $b_B \in \ell_1$ for $B \in \iota$, and $u \in \mathbb{R}^\iota$. Then for every $k \in \mathbb{N}$,*

$$\Big(V + \sum_B u_B b_B\Big)(k) = V(k) + \sum_B u_B b_B(k).$$

Proof. The vector operations on ℓ_1 are pointwise. □

Lemma 2.23 (Affine maps are weighted contractions)**.** *In the setting of Lemma 2.22, for all $u, u' \in \mathbb{R}^\iota$,*

$$\Big\| \Big(V + \sum_B u_B b_B\Big) - \Big(V + \sum_B u'_B b_B\Big) \Big\|_1 \le \sum_B \|b_B\|_1\, |u_B - u'_B|.$$

Proof. The difference is $\sum_B (u_B - u'_B) b_B$. The claim follows from the triangle inequality and homogeneity of the norm. □

Lemma 2.24 (Distances from coordinatewise cut identities)**.** *Let ι be finite, let $x_1, \dots, x_n \in \ell_1$, and let $b_B \in \ell_1$ for $B \in \iota$ with $b_B(k) \ge 0$ for all B, k. Let $z_1, \dots, z_n \in \mathbb{R}^\iota$, and assume that for all i, j, k,*

$$|x_i(k) - x_j(k)| = \sum_B |z_i(B) - z_j(B)|\, b_B(k).$$

Then $\|x_i - x_j\|_1 = \sum_B \|b_B\|_1\, |z_i(B) - z_j(B)|$ for all i, j.

Proof. By Lemma 2.20, each b_B is summable, and $\|b_B\|_1 = \sum_k b_B(k)$ because $b_B \ge 0$. Using Lemma 2.20 for $x_i - x_j$, the hypothesis, and the fact that a finite sum of convergent series may be summed termwise,

$$\|x_i - x_j\|_1 = \sum_k \sum_B |z_i(B) - z_j(B)|\, b_B(k) = \sum_B |z_i(B) - z_j(B)| \sum_k b_B(k).$$ □

Lemma 2.25 (Cut coordinates in ℓ_1)**.** *Let $x_1, \dots, x_n \in \ell_1$. There exist $b_B \in \ell_1$ for $B \in \mathcal{P}_n$ and $V \in \ell_1$ such that, for all B, i, j, k:*

1. *$b_B(k) \ge 0$;*
2. *$x_i(k) = V(k) + \sum_B \mathbf{1}[i \in B]\, b_B(k)$;*

3. $|x_i(k) - x_j(k)| = \sum_B \big|\mathbf{1}[i \in B] - \mathbf{1}[j \in B]\big|\, b_B(k)$.

Proof. Take g and v from Lemma 2.19. For each k let $a^{(k)} = (x_1(k), \dots, x_n(k))$, and define the sequences $\beta_B(k) = g_B(a^{(k)})$ and $\nu(k) = v(a^{(k)})$.

By Lemma 2.20, the sequence $k \mapsto \sum_i |x_i(k)|$ is a finite sum of summable sequences, hence summable. By Lemma 2.19,

$$|\beta_B(k)| \le 2 \sum_i |x_i(k)| \qquad \text{and} \qquad |\nu(k)| \le \sum_i |x_i(k)|.$$

So Lemma 2.21 yields $b_B = \beta_B \in \ell_1$ and $V = \nu \in \ell_1$. Properties (1)–(3) are Lemma 2.19 applied to $a^{(k)}$. □

Lemma 2.26 (Cut representation with a contractive reconstruction)**.** *Let $x_1, \dots, x_n \in \ell_1$. There exist $w \in \mathbb{R}^{\mathcal{P}_n}$ and a map $T : \mathbb{R}^{\mathcal{P}_n} \to \ell_1$ such that:*

1. *$w_B \ge 0$ for all B;*
2. *$T(\chi_i) = x_i$ for all i;*
3. *$\|x_i - x_j\|_1 = \sum_B w_B |\chi_i(B) - \chi_j(B)|$ for all i, j;*
4. *$\|T(u) - T(u')\|_1 \le \sum_B w_B |u_B - u'_B|$ for all $u, u' \in \mathbb{R}^{\mathcal{P}_n}$.*

Proof. Take b_B and V from Lemma 2.25, and put $w_B = \|b_B\|_1$ and $T(u) = V + \sum_B u_B b_B$; thus T is affine.

- By Lemma 2.22 and Lemma 2.25(2), $T(\chi_i)(k) = x_i(k)$ for every k, which gives (2).
- Property (3) is Lemma 2.24 with $z_i = \chi_i$, using Lemma 2.25(1) and (3).
- Property (4) is Lemma 2.23. □

Lemma 2.27 (Transfer from the cube to ℓ_1)**.** *Let $t \in \mathbb{N}$, let $\pi \in \mathbb{R}^n$ with $\pi \ge 0$, and let $A \in \mathbb{R}^{n \times n}$ with $a_{ij} \ge 0$. Let $x_1, \dots, x_n \in \ell_1$, let ι be finite, $w \in \mathbb{R}^\iota$ with $w \ge 0$, and $z_1, \dots, z_n \in \{0,1\}^\iota$. Let $T : \mathbb{R}^\iota \to \ell_1$ satisfy:*

- *$T(z_i) = x_i$;*
- *$\|x_i - x_j\|_1 = \|z_i - z_j\|_{w,1}$;*
- *$\|T(u) - T(u')\|_1 \le \|u - u'\|_{w,1}$ for all u, u'.*

Let $G \in \mathbb{R}^{n \times n}$ be arbitrary and let $c_1, \dots, c_n \in \mathbb{R}^\iota$ satisfy

$$\sum_i \pi_i \|z_i - c_i\|_{w,1}^2 + t \sum_{i,j} \pi_i a_{ij} \|c_i - c_j\|_{w,1}^2 \le 108 \sum_{i,j} \pi_i G_{ij} \|z_i - z_j\|_w^4.$$

Then there exist $y_1, \dots, y_n \in \ell_1$ with

$$\sum_i \pi_i \|x_i - y_i\|_1^2 + t \sum_{i,j} \pi_i a_{ij} \|y_i - y_j\|_1^2 \le 108 \sum_{i,j} \pi_i G_{ij} \|x_i - x_j\|_1^2.$$

Proof. Put $y_i = T(c_i)$. Then

$$\|x_i - y_i\|_1 = \|T(z_i) - T(c_i)\|_1 \le \|z_i - c_i\|_{w,1} \qquad \text{and} \qquad \|y_i - y_j\|_1 \le \|c_i - c_j\|_{w,1}.$$

Both sides are nonnegative, so these inequalities survive squaring. Multiplying by $\pi_i \ge 0$ and by $t\,\pi_i a_{ij} \ge 0$ and summing shows that the left side is at most the left side of the hypothesis.

For binary numbers $|\alpha - \beta| = (\alpha - \beta)^2$. Hence $\|z_i - z_j\|_w^2 = \|z_i - z_j\|_{w,1} = \|x_i - x_j\|_1$, so $\|z_i - z_j\|_w^4 = \|x_i - x_j\|_1^2$, and the right sides coincide. □

2.5 Geometric versus Cesàro averages

Lemma 2.28 (Weighted Minkowski inequality)**.** *Let α be finite and $\mu, a, b \in \mathbb{R}^\alpha$ with $\mu \geq 0$. Then*

$$\sqrt{\sum_k \mu_k (a_k + b_k)^2} \leq \sqrt{\sum_k \mu_k a_k^2} + \sqrt{\sum_k \mu_k b_k^2}.$$

Proof. This is the triangle inequality for the Euclidean norm, applied to the vectors $(\sqrt{\mu_k} a_k)_k$ and $(\sqrt{\mu_k} b_k)_k$. □

Lemma 2.29 (Weighted triangle inequality)**.** *Let α be finite and $\mu, a, b, c \in \mathbb{R}^\alpha$ with $\mu \geq 0$, $c \geq 0$ and $c_k \leq a_k + b_k$ for all k. Then*

$$\sqrt{\sum_k \mu_k c_k^2} \leq \sqrt{\sum_k \mu_k a_k^2} + \sqrt{\sum_k \mu_k b_k^2}.$$

Proof. Let α_0 and β_0 denote the two square roots on the right. Since $0 \leq c_k \leq |a_k| + |b_k|$, we have $c_k^2 \leq a_k^2 + 2|a_k||b_k| + b_k^2$. By Cauchy–Schwarz with weights $\sqrt{\mu_k}$, $\sum_k \mu_k |a_k||b_k| \leq \alpha_0 \beta_0$. Hence

$$\sum_k \mu_k c_k^2 \leq \alpha_0^2 + 2\alpha_0\beta_0 + \beta_0^2 = (\alpha_0 + \beta_0)^2.$$

Take square roots. □

Lemma 2.30 (Expanding a product of two matrices)**.** *For $\pi \in \mathbb{R}^n$ and $P, Q, d \in \mathbb{R}^{n \times n}$,*

$$\sum_{i,j} \pi_i (PQ)_{ij} d_{ij} = \sum_{(i,k,j) \in [n]^3} \pi_i P_{ik} Q_{kj} d_{ij}.$$

Proof. Expand $(PQ)_{ij} = \sum_k P_{ik} Q_{kj}$ and rearrange the finite sum. □

Lemma 2.31 (Summing out the last index)**.** *Let $\pi \in \mathbb{R}^n$ and $P, Q, d \in \mathbb{R}^{n \times n}$ with $\sum_j Q_{kj} = 1$ for all k. Then*

$$\sum_{(i,k,j)} \pi_i P_{ik} Q_{kj} d_{ik} = \sum_{i,k} \pi_i P_{ik} d_{ik}.$$

Proof. Sum over j first and use $\sum_j Q_{kj} = 1$. □

Lemma 2.32 (Summing out the first index)**.** *Let $\pi \in \mathbb{R}^n$ and $P, Q, d \in \mathbb{R}^{n \times n}$ with $\sum_i \pi_i P_{ik} = \pi_k$ for all k. Then*

$$\sum_{(i,k,j)} \pi_i P_{ik} Q_{kj} d_{kj} = \sum_{k,j} \pi_k Q_{kj} d_{kj}.$$

Proof. Sum over i first and use $\sum_i \pi_i P_{ik} = \pi_k$. □

Lemma 2.33 (Square-root subadditivity of the endpoint energy)**.** *Let π be a probability vector, A stochastic with $\pi A = \pi$, and $x_1, \ldots, x_n \in \ell_1$. Put*

$$D(s) = \sum_{i,j} \pi_i (A^s)_{ij} \|x_i - x_j\|_1^2.$$

Then $\sqrt{D(r+s)} \leq \sqrt{D(r)} + \sqrt{D(s)}$ for all $r, s \in \mathbb{N}$.

Proof. Let $\mu(i,k,j) = \pi_i (A^r)_{ik} (A^s)_{kj}$; it is nonnegative by Lemma 2.2. We express all three quantities as μ-weighted sums.

- Since $A^{r+s} = A^r A^s$, Lemma 2.30 gives $D(r+s) = \sum \mu(i,k,j)\|x_i - x_j\|_1^2$.
- Since the rows of A^s sum to 1 (Lemma 2.2), Lemma 2.31 gives $D(r) = \sum \mu(i,k,j)\|x_i - x_k\|_1^2$.
- Since $\pi A^r = \pi$ (Lemma 2.3), Lemma 2.32 gives $D(s) = \sum \mu(i,k,j)\|x_k - x_j\|_1^2$.

The triangle inequality $\|x_i - x_j\|_1 \le \|x_i - x_k\|_1 + \|x_k - x_j\|_1$ and Lemma 2.29 conclude. □

Lemma 2.34 (Bound on $D(t)$)**.** *Let $t \ge 1$ and let $D : \mathbb{N} \to \mathbb{R}$ satisfy $D(0) = 0$, $D \ge 0$ and $\sqrt{D(r+s)} \le \sqrt{D(r)} + \sqrt{D(s)}$ for all r, s. Then*

$$D(t) \le 4 \cdot \frac{1}{t} \sum_{s=1}^{t} D(s).$$

Proof. For $1 \le r \le t$, squaring $\sqrt{D(t)} \le \sqrt{D(r)} + \sqrt{D(t-r)}$ and using $(\alpha+\beta)^2 \le 2\alpha^2 + 2\beta^2$ gives $D(t) \le 2D(r) + 2D(t-r)$. Sum over $r = 1, \dots, t$. As r runs over $1, \dots, t$, the index $t - r$ runs over $0, \dots, t-1$, and $D(0) = 0$. Hence

$$t\, D(t) \le 2 \sum_{r=1}^{t} D(r) + 2 \sum_{s=1}^{t-1} D(s) \le 4 \sum_{s=1}^{t} D(s),$$

using $D \ge 0$ in the last step. Divide by t. □

Lemma 2.35 (Block decomposition)**.** *Let $t \in \mathbb{N}$ and let $D : \mathbb{N} \to \mathbb{R}$ satisfy $D \ge 0$ and $\sqrt{D(r+s)} \le \sqrt{D(r)} + \sqrt{D(s)}$ for all r, s. Then for all $j, r \in \mathbb{N}$,*

$$D(jt + r) \le 2j^2 D(t) + 2D(r).$$

Proof. By induction on j, using $(j+1)t + r = (jt + r) + t$ and subadditivity, we get $\sqrt{D(jt+r)} \le j\sqrt{D(t)} + \sqrt{D(r)}$. Squaring (both sides are nonnegative) and using $(\alpha+\beta)^2 \le 2\alpha^2 + 2\beta^2$ gives the claim. □

Lemma 2.36 (Second moment of the geometric series)**.** *For $0 \le r < 1$, the series $\sum_{s \ge 0} s^2 r^s$ converges, and its sum is $r(1+r)/(1-r)^3$.*

Proof. We have $s^2 = 2\binom{s+2}{2} - 3s - 2$. We also have the standard sums

$$\sum_s \binom{s+2}{2} r^s = \frac{1}{(1-r)^3}, \qquad \sum_s s r^s = \frac{r}{(1-r)^2}, \qquad \sum_s r^s = \frac{1}{1-r},$$

all of which converge. Combining them linearly,

$$\frac{2}{(1-r)^3} - \frac{3r}{(1-r)^2} - \frac{2}{1-r} = \frac{2 - 3r + 3r^2 - 2 + 4r - 2r^2}{(1-r)^3} = \frac{r(1+r)}{(1-r)^3}.$$

□

Lemma 2.37 (Floor bound)**.** *For $t \ge 1$ and $s \in \mathbb{N}$, $\lfloor s/t \rfloor^2 \le s^2/t^2$.*

Proof. We have $t\lfloor s/t \rfloor \le s$, so $0 \le \lfloor s/t \rfloor \le s/t$. Square. □

Lemma 2.38 (Geometric second moment of the block index)**.** *Let $t \ge 1$. Then the series $\sum_{s \ge 0} p\, q^s \lfloor s/t \rfloor^2$ converges and its sum is at most 3.*

Proof. By Lemma 2.37, the terms are nonnegative and bounded by $\frac{p}{t^2}s^2q^s$. By Lemma 2.36, these bounds are summable with sum

$$\frac{p}{t^2}\cdot\frac{q(1+q)}{(1-q)^3}.$$

Using $1-q=1/(t+1)$ and $q=t/(t+1)$, this equals $(2t+1)/t\le 3$. □

Lemma 2.39 (Geometric weight of a residue class)**.** *Let $t\ge 1$ and $r,N\in\mathbb{N}$. Then*

$$\sum_{\substack{0\le s<N\\ s \bmod t=r}} p\,q^s \le \frac{2}{t}.$$

Proof. The factor q^t. We have $q(1+1/t)=1$, and by Bernoulli's inequality $(1+1/t)^t\ge 1+t\cdot\frac{1}{t}=2$. Hence $q^t\le 1/2$.

Termwise bound. For s in the sum, $q^s=q^{s \bmod t}(q^t)^{\lfloor s/t\rfloor}\le(1/2)^{\lfloor s/t\rfloor}$, because $0\le q\le 1$.

Reindexing. The map $s\mapsto\lfloor s/t\rfloor$ is injective on the residue class, since $s=(s \bmod t)+t\lfloor s/t\rfloor$. It maps $\{0,\dots,N-1\}$ into itself, since $\lfloor s/t\rfloor\le s$. Therefore the sum is at most

$$p\sum_{k<N}2^{-k}=p\,(2-2\cdot 2^{-N})\le 2p\le\frac{2}{t}.$$ □

Lemma 2.40 (Geometric average of a periodic function)**.** *Let $t\ge 1$ and $f:\mathbb{N}\to\mathbb{R}$ with $f\ge 0$. Then the series $\sum_{s\ge 0}p\,q^s f(s \bmod t)$ converges, and its sum is at most $\frac{2}{t}\sum_{r=0}^{t-1}f(r)$.*

Proof. Convergence. The terms are nonnegative and bounded by $p\,q^s\sum_{r<t}f(r)$, which is summable. So the series converges.

Bound. For every N, group the partial sum by the residue $r=s \bmod t\in\{0,\dots,t-1\}$ and apply Lemma 2.39:

$$\sum_{s<N}p\,q^s f(s \bmod t)=\sum_{r<t}f(r)\sum_{\substack{s<N\\ s \bmod t=r}}p\,q^s\le\frac{2}{t}\sum_{r<t}f(r).$$

A series of nonnegative terms whose partial sums are bounded by a constant has sum at most that constant. □

Lemma 2.41 (Geometric averages of bounded sequences)**.** *Let $t\in\mathbb{N}$ and let $f:\mathbb{N}\to\mathbb{R}$ satisfy $f\ge 0$ and $f\le C$ for some constant C. Then the series $\sum_{s\ge 0}p\,q^s f(s)$ converges.*

Proof. We have $0\le p\,q^s f(s)\le C\,p\,q^s$, and $\sum_s q^s$ converges because $0\le q<1$. □

Lemma 2.42 (Euclidean-division bound)**.** *Let $t\in\mathbb{N}$ and let $D:\mathbb{N}\to\mathbb{R}$ satisfy $D\ge 0$ and $\sqrt{D(r+s)}\le\sqrt{D(r)}+\sqrt{D(s)}$ for all r,s. Then for every $s\in\mathbb{N}$,*

$$D(s)\le 2\lfloor s/t\rfloor^2 D(t)+2D(s \bmod t).$$

For $t=0$ this is read with the conventions $\lfloor s/0\rfloor=0$ and $s \bmod 0=s$.

Proof. If $t=0$, the claim reads $D(s)\le 2D(s)$, which holds since $D\ge 0$.

Let $t\ge 1$, and write $m=\lfloor s/t\rfloor$ and $\rho=s \bmod t$, so that $s=mt+\rho$. The formal proof does not cite Lemma 2.35; it reproduces the same bound by the following route.

1. *Two-term bound.* For all $a,b\in\mathbb{N}$, squaring $\sqrt{D(a+b)}\le\sqrt{D(a)}+\sqrt{D(b)}$ (both sides are nonnegative) and using $(\alpha+\beta)^2\le 2\alpha^2+2\beta^2$ gives $D(a+b)\le 2D(a)+2D(b)$.

2. *Multiples of t.* For $m \geq 1$, we have $\sqrt{D(mt)} \leq m\sqrt{D(t)}$, by induction on m. The case $m = 1$ is trivial. For the induction step, $\sqrt{D((m+1)t)} \leq \sqrt{D(mt)} + \sqrt{D(t)} \leq (m+1)\sqrt{D(t)}$. Squaring gives $D(mt) \leq m^2 D(t)$.

3. *The case $m = 0$.* Then $s = \rho$, and $D(s) = D(\rho) \leq 2D(\rho) = 2 \cdot 0^2 \cdot D(t) + 2D(\rho)$, since $D \geq 0$.

4. *The case $m \geq 1$.* By steps 1 and 2,

$$D(s) = D(mt + \rho) \leq 2D(mt) + 2D(\rho) \leq 2m^2 D(t) + 2D(\rho).$$

This is the bound of Lemma 2.35 with $j = m$ and $r = \rho$. □

Lemma 2.43 (Splitting the geometric average)**.** *Let $t \geq 1$ and let $D : \mathbb{N} \to \mathbb{R}$ be nonnegative, bounded, and square-root subadditive as in Lemma 2.42. Then*

$$\sum_{s \geq 0} p\, q^s D(s) \leq 2D(t) \sum_{s \geq 0} p\, q^s \lfloor s/t \rfloor^2 + 2 \sum_{s \geq 0} p\, q^s D(s \bmod t).$$

Proof. Convergence. The series on the left and the last series converge, since D is bounded and nonnegative (compare with $C\, p\, q^s$). The middle series converges because $0 \leq \lfloor s/t \rfloor^2 q^s \leq s^2 q^s$, which is summable by Lemma 2.36.

Termwise bound. Multiplying Lemma 2.42 by $p\, q^s \geq 0$ bounds the series termwise. Comparison of convergent series and linearity of the sum give the claim. □

Lemma 2.44 (Shifting a Cesàro window)**.** *Let $t \in \mathbb{N}$ and let $D : \mathbb{N} \to \mathbb{R}$ satisfy $D(0) = 0$ and $D \geq 0$. Then*

$$\sum_{r=0}^{t-1} D(r) \leq \sum_{s=1}^{t} D(s).$$

Proof. For $t = 0$ both sides vanish. For $t \geq 1$, the left side equals $\sum_{r=1}^{t-1} D(r)$ because $D(0) = 0$, and this is at most the same sum plus $D(t) \geq 0$. □

Lemma 2.45 (Geometric versus Cesàro averages)**.** *Let $t \geq 1$ and let $D : \mathbb{N} \to \mathbb{R}$ satisfy $D(0) = 0$, $D \geq 0$, D bounded, and $\sqrt{D(r+s)} \leq \sqrt{D(r)} + \sqrt{D(s)}$ for all r, s. Then*

$$\sum_{s \geq 0} p\, q^s D(s) \leq 28 \cdot \frac{1}{t} \sum_{s=1}^{t} D(s).$$

Proof. Let $W = \frac{1}{t} \sum_{s=1}^{t} D(s)$. By Lemma 2.43, the left side is at most $2D(t) \cdot S_2 + 2M$, where $S_2 = \sum_s p\, q^s \lfloor s/t \rfloor^2$ and $M = \sum_s p\, q^s D(s \bmod t)$. We bound each piece.

- $S_2 \leq 3$ by Lemma 2.38; since $D(t) \geq 0$, this gives $2D(t)S_2 \leq 6D(t)$.
- $M \leq \frac{2}{t} \sum_{r<t} D(r) \leq 2W$ by Lemma 2.40 and Lemma 2.44.
- $D(t) \leq 4W$ by Lemma 2.34.

Altogether the left side is at most $6 \cdot 4W + 2 \cdot 2W = 28W$. □

Lemma 2.46 (Resolvent energy as a geometric average)**.** *Let $t \in \mathbb{N}$, $\pi \in \mathbb{R}^n$, A stochastic, $x_1, \ldots, x_n \in \ell_1$, and let G be the resolvent matrix of A. With $D(s) = \sum_{i,j} \pi_i (A^s)_{ij} \|x_i - x_j\|_1^2$,*

$$\sum_{i,j} \pi_i G_{ij} \|x_i - x_j\|_1^2 = \sum_{s \geq 0} p\, q^s D(s).$$

Proof. Each series $G_{ij} = \sum_s p\,q^s (A^s)_{ij}$ converges by Lemma 2.4. Multiply it by $\pi_i \|x_i - x_j\|_1^2$ and add these finitely many convergent series. This shows that $\sum_s p\,q^s D(s)$ converges, and that its sum is the left side. □

Lemma 2.47 (Properties of the endpoint energy)**.** *Let π be a probability vector, A stochastic, $x_1, \dots, x_n \in \ell_1$, and D as in Lemma 2.46. Then:*

1. *$D(0) = 0$;*
2. *$D(s) \geq 0$ for all s;*
3. *there is a constant C with $D(s) \leq C$ for all s.*

Proof. 1. Since $A^0 = I$, $D(0) = \sum_i \pi_i \|x_i - x_i\|_1^2 = 0$.

2. Nonnegativity follows from $\pi \geq 0$ and $(A^s)_{ij} \geq 0$ (Lemma 2.2).

3. Each entry of A^s is nonnegative and at most its row sum 1, and $\pi_i \leq \sum_k \pi_k = 1$. Hence $D(s) \leq C := \sum_{i,j} \|x_i - x_j\|_1^2$. □

Lemma 2.48 (Cesàro energy)**.** *Let $t \in \mathbb{N}$, $\pi \in \mathbb{R}^n$, $A \in \mathbb{R}^{n \times n}$, $x_1, \dots, x_n \in \ell_1$, and D as above. Then*

$$\sum_{i,j} \pi_i \Big(\frac{1}{t} \sum_{s=1}^{t} (A^s)_{ij} \Big) \|x_i - x_j\|_1^2 = \frac{1}{t} \sum_{s=1}^{t} D(s).$$

For $t = 0$ both sides vanish, since the inner sums are empty.

Proof. Pull out the factor $1/t$ and exchange the finite sums over s and over (i, j). □

Lemma 2.49 (Comparison of resolvent and Cesàro energies)**.** *Let $t \geq 1$, let π be a probability vector, let A be stochastic with $\pi A = \pi$, let $x_1, \dots, x_n \in \ell_1$, and let G be the resolvent matrix of A. Then*

$$\sum_{i,j} \pi_i G_{ij} \|x_i - x_j\|_1^2 \leq 28 \sum_{i,j} \pi_i \Big(\frac{1}{t} \sum_{s=1}^{t} (A^s)_{ij} \Big) \|x_i - x_j\|_1^2.$$

Proof. Let $D(s) = \sum_{i,j} \pi_i (A^s)_{ij} \|x_i - x_j\|_1^2$.

- By Lemma 2.46, the left side equals $\sum_s p\,q^s D(s)$.
- By Lemma 2.48, the sum on the right equals $\frac{1}{t} \sum_{s=1}^{t} D(s)$.
- By Lemma 2.47 and Lemma 2.33, D satisfies the hypotheses of Lemma 2.45, which gives the bound. □

3 Proof of the theorem

Proof of Theorem 1.3. Let $n, t, \pi, A, x_1, \dots, x_n$ be as in the theorem, and let $p = 1/(t+1)$ and $q = t/(t+1)$.

Step 1: stationarity and cut representation. By Lemma 2.1, $\pi A = \pi$. By Lemma 2.26, there are weights $w \in \mathbb{R}^{\mathcal{P}_n}$ with $w \geq 0$ and a map $T : \mathbb{R}^{\mathcal{P}_n} \to \ell_1$ with $T(\chi_i) = x_i$, $\|x_i - x_j\|_1 = \|\chi_i - \chi_j\|_{w,1}$, and $\|T(u) - T(u')\|_1 \leq \|u - u'\|_{w,1}$.

Step 2: the resolvent. Let G be the resolvent matrix, $G_{ij} = \sum_{s \geq 0} p\,q^s (A^s)_{ij}$. By Lemma 2.5, $G_{ij} \geq 0$, $\sum_j G_{ij} = 1$ and $\sum_i \pi_i G_{ij} = \pi_j$. By Lemma 2.6, $G = pI + qAG = pI + qGA$.

Step 3: the energy estimate on the cube. Take $\iota = \mathcal{P}_n$ and $z_i = \chi_i \in \{0,1\}^{\mathcal{P}_n}$, and let $h_i = \sum_j G_{ij}\chi_j$ and $c_i = \Phi(h_i)$. By Lemma 2.16,

$$\sum_i \pi_i \|\chi_i - c_i\|_{w,1}^2 + t\sum_{i,j} \pi_i a_{ij}\|c_i - c_j\|_{w,1}^2 \le 108 \sum_{i,j} \pi_i G_{ij}\|\chi_i - \chi_j\|_w^4.$$

Step 4: transfer to ℓ_1. The hypotheses of Lemma 2.27 hold, since $\pi \ge 0$ and $a_{ij} \ge 0$. It yields points $y_i = T(\Phi(h_i)) \in \ell_1$ with

$$\sum_i \pi_i \|x_i - y_i\|_1^2 + t\sum_{i,j} \pi_i a_{ij}\|y_i - y_j\|_1^2 \le 108 \sum_{i,j} \pi_i G_{ij}\|x_i - x_j\|_1^2.$$

Step 5: comparison. By Lemma 2.49, which uses $t \ge 1$ and $\pi A = \pi$, the right side is at most

$$108 \cdot 28 \sum_{i,j} \pi_i \Big(\frac{1}{t}\sum_{s=1}^{t} (A^s)_{ij}\Big)\|x_i - x_j\|_1^2.$$

Since $108 \cdot 28 = 3024$, this proves the theorem. □

Relation to the source proof. The constructed points are the same as in the source: $y_i = T(\Phi(h_i))$, where h_i is the resolvent average of the binary cut encodings. The constants are also the same: 108 from the flatness lemma and 28 from the geometric/Cesàro comparison. The formal argument deviates from the source in the following respects.

- *No martingale.* The source realises the cost through a killed Markov chain and an H-valued martingale, and invokes conditional expectations and convergence theorems. The formal proof replaces this with deterministic algebra.
  - The martingale property of one step is the first-step identity $h_j = p\, z_j + q\sum_k a_{jk}h_k$. This identity is what kills the linear term in Lemma 2.11.
  - The summation over time against the occupation measure of the killed chain becomes a weighting by $\pi_i G_{ij}$, which telescopes through $G = pI + qGA$ in Lemma 2.12.
  - Discarding the initial potential corresponds to dropping the nonnegative terms $p\,\pi_i H_{ii}$.
  - The identity $B + tE_A$ becomes the final multiplication by $t + 1$.
- *Comparison lemma.* The comparison of geometric and Cesàro averages is proved analytically for the sequence $D(s)$, and only for points of ℓ_1. There is no random time S. The bounds $\mathbb{E}J^2 \le 3$ and $\mathbb{P}(R = r) \le 2/t$ of the source appear as Lemma 2.38 and Lemma 2.39.
- *Cut decomposition.* The decomposition uses all 2^n subsets of $[n]$ with weights $w_B \ge 0$. Zero-weight cuts are not discarded, and there is no separate case for coinciding points. The cut vectors are obtained by sorting each coordinate and making a choice, rather than by the explicit min–max formula, though the resulting nested cut structure is the same.
- *Reversibility.* Reversibility is used only to derive stationarity (Lemma 2.1). Every later step uses only $\pi A = \pi$, in agreement with the remark in the source.

A Correspondence with the formal proof

Each lemma of the text and the Lean declaration that states it; Theorem 1.3 is `openaimath_332_metricmarkovcotypel1`

Lemma	Lean declaration	Status
Lemma 2.1	`mc_stationary`	verified
Lemma 2.2	`mc_pow_stochastic`	verified
Lemma 2.3	`mc_pow_stationary`	verified
Lemma 2.4	`mc_resolvent_summable`	verified
Lemma 2.5	`mc_resolvent_basic`	verified
Lemma 2.6	`mc_resolvent_first_step`	verified
Lemma 2.7	`mc_harmonic_point`	verified
Lemma 2.8	`mc_phi_scalar`	verified
Lemma 2.9	`mc_flat_increment`	verified
Lemma 2.10	`mc_flat`	verified
Lemma 2.11	`mc_one_step`	verified
Lemma 2.12	`mc_aggregate`	verified
Lemma 2.13	`mc_phi_vertex`	verified
Lemma 2.14	`mc_cube_step`	verified
Lemma 2.15	`mc_cube_energy_raw`	verified
Lemma 2.16	`mc_cube_energy`	verified
Lemma 2.17	`mc_cut_real_mono`	verified
Lemma 2.18	`mc_cut_real_pointwise`	verified
Lemma 2.19	`mc_cut_real`	verified
Lemma 2.20	`mc_L1_norm`	verified
Lemma 2.21	`mc_L1_of_bound`	verified
Lemma 2.22	`mc_L1_affine_apply`	verified
Lemma 2.23	`mc_L1_affine_contr`	verified
Lemma 2.24	`mc_L1_dist_eq`	verified
Lemma 2.25	`mc_cut_coords`	verified
Lemma 2.26	`mc_cut_L1`	verified
Lemma 2.27	`mc_transfer`	verified
Lemma 2.28	`mc_minkowski`	verified
Lemma 2.29	`mc_weighted_tri`	verified
Lemma 2.30	`mc_triple_mul`	verified
Lemma 2.31	`mc_triple_first`	verified
Lemma 2.32	`mc_triple_second`	verified
Lemma 2.33	`mc_D_subadd`	verified
Lemma 2.34	`mc_cesaro_Dt`	verified
Lemma 2.35	`mc_cesaro_blocks`	verified
Lemma 2.36	`mc_hasSum_sq_geom`	verified
Lemma 2.37	`mc_floor_div_sq_le`	verified
Lemma 2.38	`mc_geom_sq_div`	verified
Lemma 2.39	`mc_geom_mod_coeff`	verified
Lemma 2.40	`mc_geom_mod`	verified
Lemma 2.41	`mc_geom_summable_bdd`	verified
Lemma 2.42	`mc_D_divmod`	verified

Lemma	Lean declaration	Status
Lemma 2.43	`mc_geom_cesaro_pre`	verified
Lemma 2.44	`mc_range_le_Icc`	verified
Lemma 2.45	`mc_geom_cesaro`	verified
Lemma 2.46	`mc_resolvent_D_eq`	verified
Lemma 2.47	`mc_D_props`	verified
Lemma 2.48	`mc_cesaro_rhs_eq`	verified
Lemma 2.49	`mc_comparison`	verified